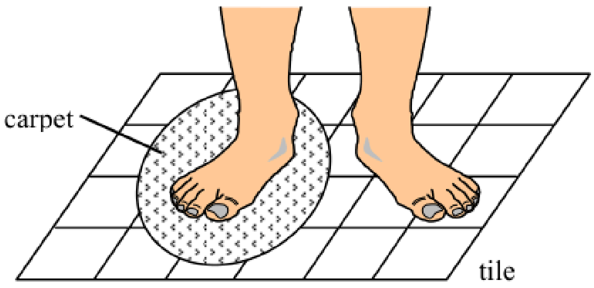
Q1

1.

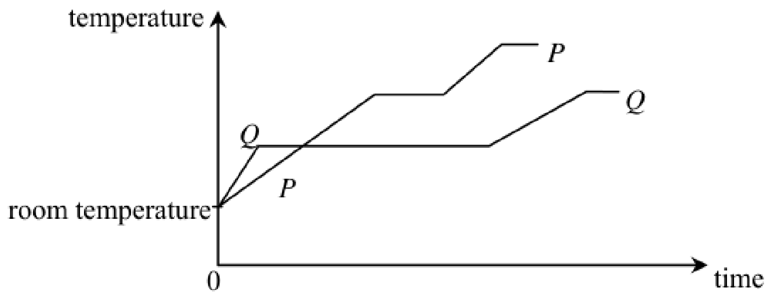


Cynthia places a piece of carpet on a tiled floor. After a while, she stands in bare feet with one foot on the tiled floor and the other on the carpet as shown above. She feels that the tiled floor is colder than the carpet. Which of the following best explains this phenomenon ?

A. The tile is a better insulator of heat than the carpet.
B. The tile is at a lower temperature than the carpet.
C. The specific heat capacity of the tile is smaller than that of the carpet.
D. Energy transfers from Cynthia's foot to the tile at a greater rate than that to the carpet.

Q2

2.

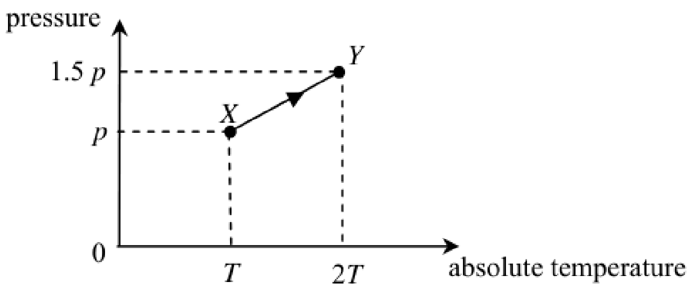


The graph shows the variation in temperature of equal masses of two substances P and Q when they are separately heated by identical heaters. Which deduction is correct ?

A. The melting point of P is lower than that of Q.
B. The specific heat capacity of P in solid state is larger than that of Q.
C. The specific latent heat of fusion of P is larger than that of Q.
D. The energy required to raise the temperature of P from room temperature to boiling point is more than that of Q.

Q3

*3.



As the gas in a vessel of fixed volume is heated, it gradually leaks out. The gas in the vessel changes from state X to state Y along the path XY shown in the plot of pressure against absolute temperature. What percentage of the original mass of the gas leaks out from the vessel in this process ?

A. 10%
B. 20%
C. 25%
D. 50%

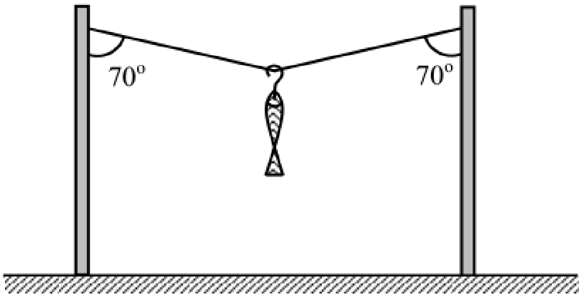
Q4

*4. Two vessels contain hydrogen gas and oxygen gas respectively. Both gases have the same pressure and temperature and are assumed to be ideal. Which of the following physical quantities must be the same for the two gases ?

A. The volume of the gas
B. The mass per unit volume of the gas
C. The r.m.s. speed of the gas molecules
D. The number of gas molecules per unit volume

Q5

5.

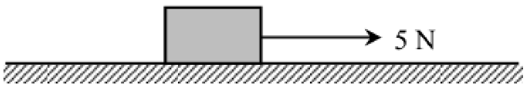


A fish is hung on a light string as shown above. The tension in the string is 10 N. Find the total weight of the fish and the hook.

A. 20 sin 70° N
B. 20 cos 70° N
C. 10 sin 70° N
D. 10 cos 70° N

Q6

6.

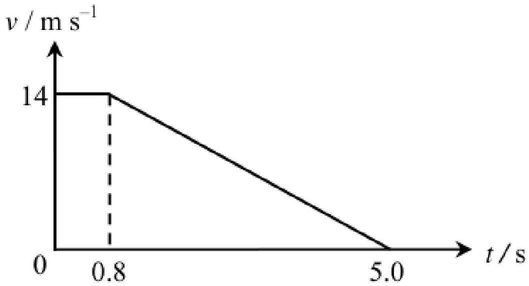


A 1 kg block is pulled by a horizontal force of 5 N and moves with an acceleration of 2 m s^{-2} on a rough horizontal plane. Find the frictional force acting on the block.

A. zero
B. 2 N
C. 3 N
D. 7 N

Q7

7. Patrick is driving along a straight horizontal road. At time $t = 0$, he observes that an accident has happened. He then applies the brakes to stop his car with uniform deceleration. The graph shows the variation of the speed of the car with time.

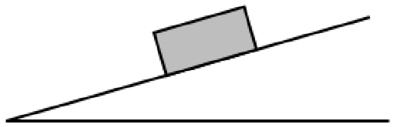


Find the distance travelled by the car from time $t = 0$ to 5.0 s.

A. 29.4 m
B. 40.6 m
C. 46.2 m
D. 81.2 m

Q8

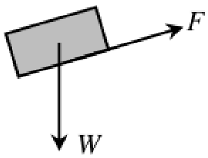
8.



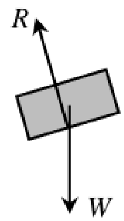
A block remains at rest on a rough inclined plane. Which diagram shows all the forces acting on the block ?

Note : W = gravitational force acting on the block,
R = normal reaction exerted by the inclined plane on the block, and
F = friction acting on the block.

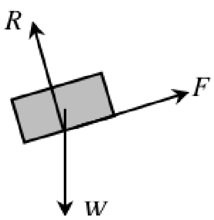
A.



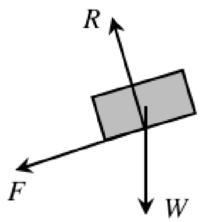
B.



C.



D.



Q9

9. Kelvin is standing on a balance inside a lift. The table shows the readings of the balance in three situations.

Motion of the lift	**Reading of the balance**
moving upwards with a uniform speed	R_1
moving downwards with a uniform speed	R_2
moving upwards with an acceleration	R_3

Which relationship is correct ?

A. $R_1 = R_2 > R_3$
B. $R_3 > R_1 = R_2$
C. $R_1 > R_2 > R_3$
D. $R_3 > R_1 > R_2$

Q10

10.

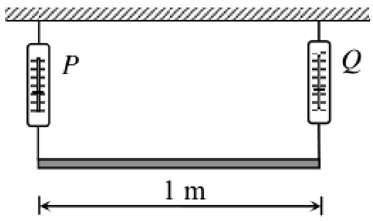


Figure (a)

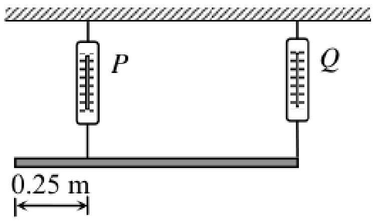


Figure (b)

Figure (a) shows a uniform plank supported by two spring balances P and Q. The readings of the two balances are both 150 N. P is now moved 0.25 m towards Q (see Figure (b)). Find the new readings of P and Q.

	Reading of P/N	**Reading of Q/N**
A.	100	200
B.	150	150
C.	200	100
D.	200	150

Q11

11. Which of the following pairs of forces is/are example(s) of action and reaction ?

(1) The centripetal force keeping a satellite in orbit round the earth and the weight of the satellite.
(2) The air resistance acting on an object falling through the air with terminal velocity and the weight of the object.
(3) The forces of attraction experienced by two parallel wires carrying currents in the same direction.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q12

12. Two small identical objects P and Q are released from rest from the top of a building 80 m above the ground. Q is released 1 s after P. Neglecting air resistance, what is the maximum vertical separation between P and Q in the air ?

A. 5 m
B. 10 m
C. 35 m
D. 45 m

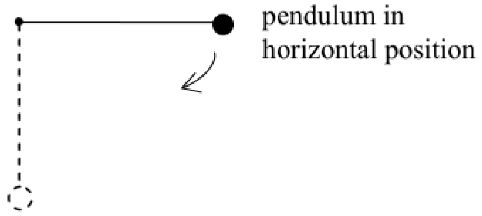
Q13

13. A car P of mass 1000 kg moves with a speed of 20 m s^{-1} and makes a head-on collision with a car Q of mass 1500 kg, which was moving with a speed of 10 m s^{-1} in the opposite direction before the collision. The two cars stick together after the collision. Find their common velocity immediately after the collision.

A. 2 m s^{-1} along the original direction of P
B. 2 m s^{-1} along the original direction of Q
C. 14 m s^{-1} along the original direction of P
D. 14 m s^{-1} along the original direction of Q

Q14

*14.

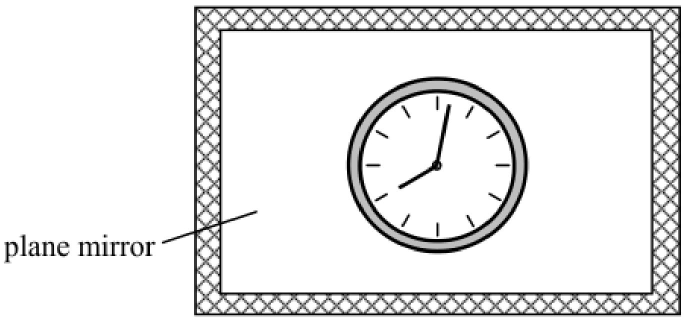


A simple pendulum is held at rest in a horizontal position. It is then released with the string taut. Which statement about the tension in the string is **not correct** when the pendulum reaches its vertical position ?

A. The tension equals the weight of the pendulum bob in magnitude.
B. The tension attains its greatest value.
C. The tension does not depend on the length of the pendulum.
D. The tension depends on the mass of the pendulum bob.

Q15

15.



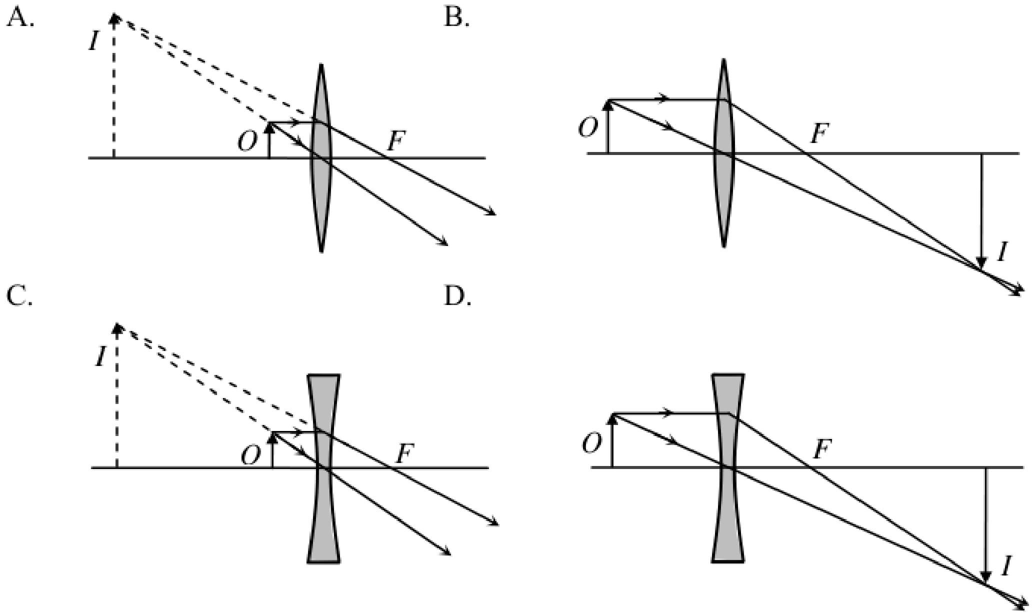
The diagram shows the image of a clock in a plane mirror. What is the time displayed by the clock ?

A. 3:58
B. 4:02
C. 7:58
D. 8:02

Q16

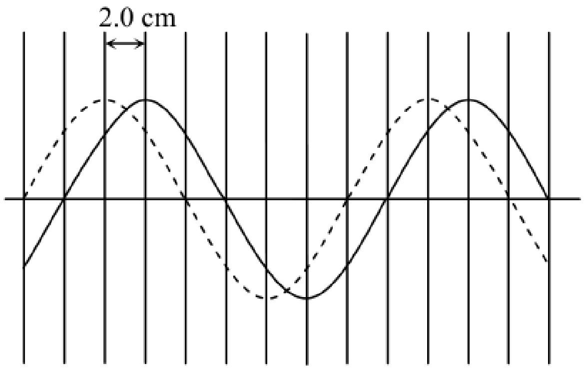
16.

Cecilia uses a magnifying glass to read some small print. Which diagram shows how the image of tl print is formed ?



Q17

17.

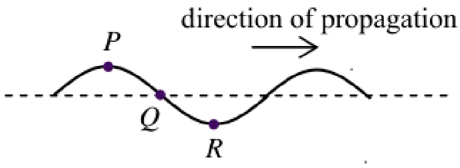


The solid curve in the diagram shows a transverse wave at a certain instant. After 0.05 s, the wave has travelled a distance of 2.0 cm and is indicated by the dashed curve. Find the wavelength and frequency of the wave.

	Wavelength/cm	**Frequency/Hz**
A.	8	2.5
B.	16	2.5
C.	8	5
D.	16	5

Q18

18.

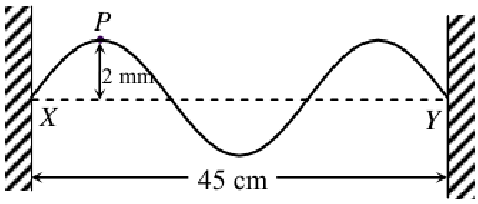


The figure shows the shape of a transverse wave travelling along a string at a certain instant. Which statement about the motion of the particles *P*, *Q* and *R* on the string at this instant is correct ?

A. Particle *P* is moving downwards.
B. Particle *Q* is stationary.
C. Particle *R* attains its maximum acceleration.
D. *P* and *Q* are in phase.

Q19

19.



String *XY* is fixed at both ends. The distance between *X* and *Y* is 45 cm. Two identical sinusoidal waves travel along *XY* in opposite directions and form a stationary wave with an antinode at point *P*. The figure shows the string when *P* is 2 mm, its maximum displacement, from the equilibrium position. What is the amplitude and wavelength of each of the **travelling waves** on the string ?

	Amplitude	**Wavelength**
A.	1 mm	30 cm
B.	1 mm	15 cm
C.	2 mm	30 cm
D.	2 mm	15 cm

Q20

20. A Young's double-slit experiment was performed using a monochromatic light source. Which change would result in a greater fringe separation on the screen ?

(1) Using monochromatic light source of longer wavelength
(2) Using double slit with greater slit separation
(3) Using double slit with larger slit width

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

Q21

21. An object is placed at the focus of a concave lens of focal length 10 cm. What is the magnification of the image formed ?

A. 0.5
B. 1.0
C. 2.0
D. infinite

Q22

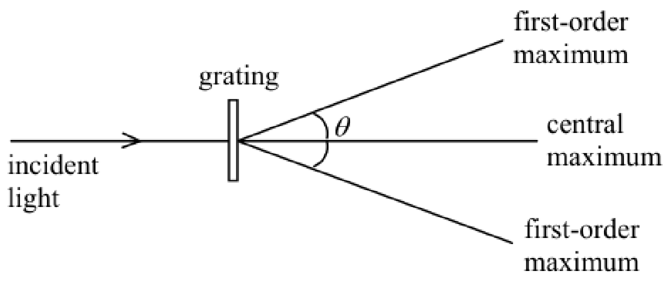
22. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are longitudinal waves.
(2) Sound waves are electromagnetic waves.
(3) Sound waves cannot travel in a vacuum.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

Q23

23.



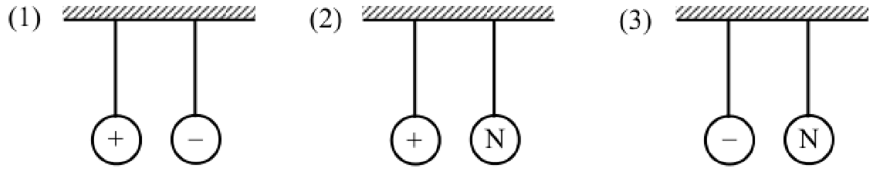
When monochromatic light is passed through a diffraction grating, a pattern of maxima and minima is observed as shown. Which combination would produce the largest angle θ between the first-order maxima ?

	Grating (lines per mm)	**Colour of light used**
A.	200	blue
B.	200	red
C.	400	blue
D.	400	red

Q24

24. Two conducting spheres are hanging freely in air by insulating threads. In which of the following will the two spheres attract each other ?

Note : 'N' denotes that the sphere is uncharged.



A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

Q25

25. The table shows three electrical appliances which Clara used in a certain month :

Appliance	Rating	Duration
Air-conditioner	220 V, 1200 W	250 hours
television	220 V, 250 W	80 hours
computer	220 V, 150 W	60 hours

Calculate the cost of electricity used.
Note : 1 kW h of electricity costs $ 0.86.

A. $ 62.25
B. $ 73.79
C. $ 282.94
D. $ 536.64

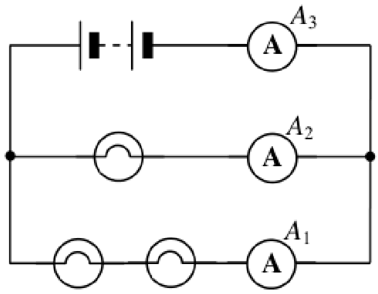
Q26

26. If a 15 A fuse is installed in the plug of an electric kettle of rating '220 V, 900 W', state what happens when the kettle is plugged in and switched on.

A. The kettle will not operate.
B. The kettle will be short-circuited.
C. The output power of the kettle will be increased.
D. The chance of the kettle being damaged by an excessive current will be increase

Q27

27.

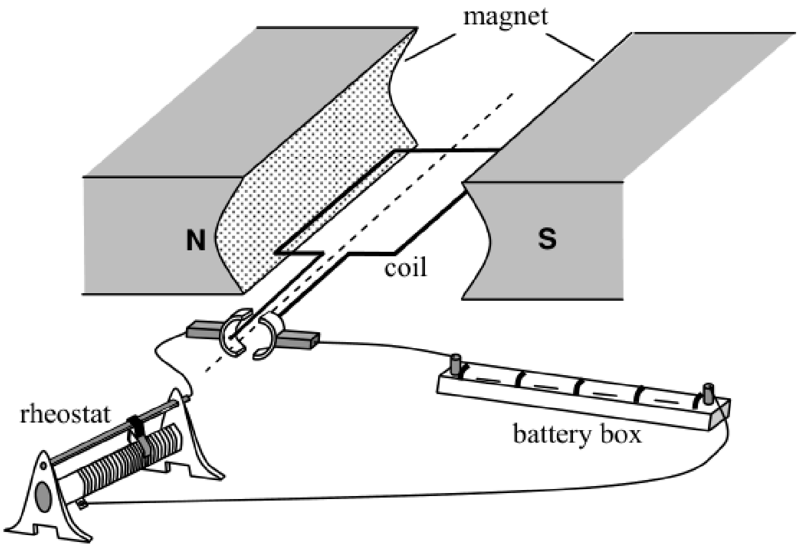


In the above circuit, the bulbs are identical. The reading of ammeter A_1 is 1 A. Find the readings of ammeters A_2 and A_3.

	Reading of A_2	**Reading of A_3**
A.	2 A	2 A
B.	2 A	3 A

Q28

28.



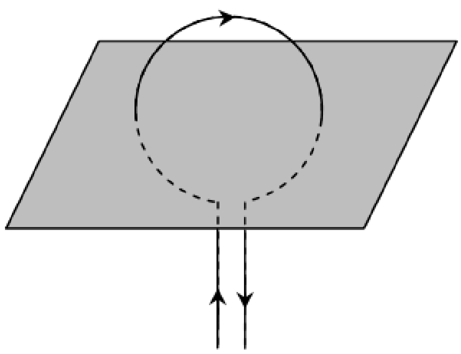
The figure shows a simple motor. Which of these changes would increase the turning effect of the coil ?

(1) using a stronger magnet
(2) reducing the resistance of the rheostat
(3) using a coil with a smaller number of turns

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

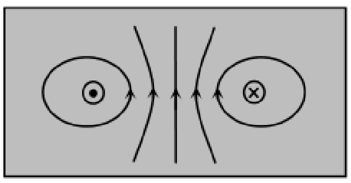
Q29

29.

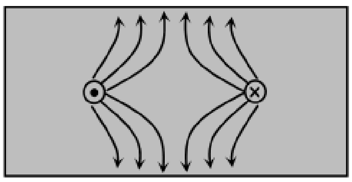


Which diagram shows the magnetic field pattern around a flat circular current-carrying coil, in the plane shown ?

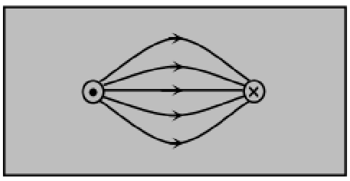
A.



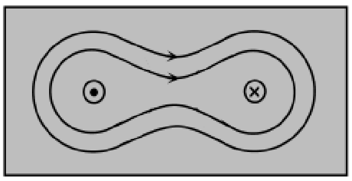
B.



C.

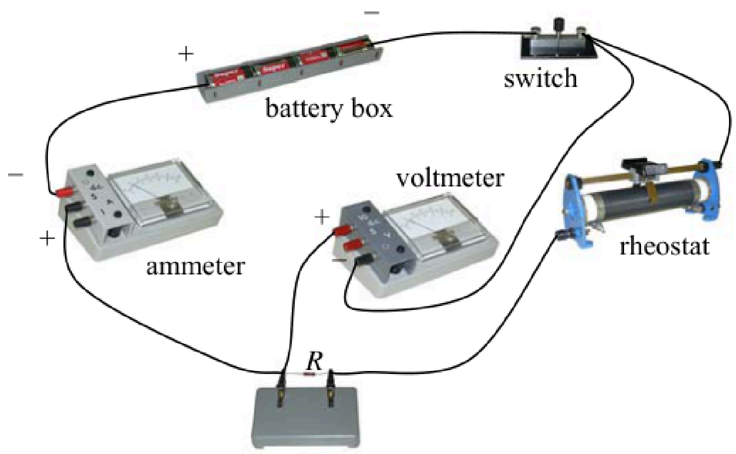


D.



Q30

30.



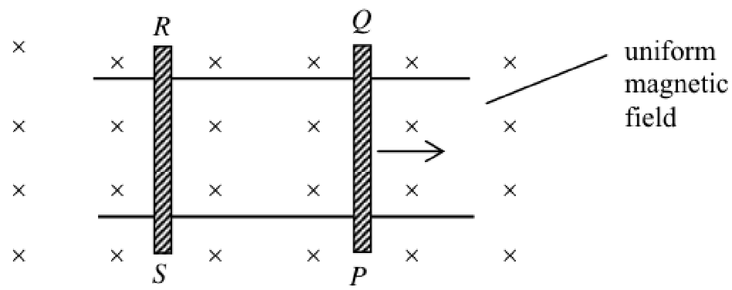
A student wants to measure the resistance of a resistor R and sets up the circuit shown. The student made which of these mistakes setting up the circuit ?

(1) The polarity of the ammeter was reversed.
(2) The polarity of the voltmeter was reversed.
(3) The voltmeter was connected across both R and the rheostat.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

Q31

31.

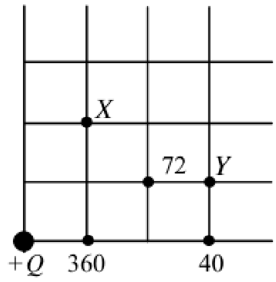


The figure shows conducting rods PQ and RS placed on two smooth, parallel, horizontal conducting rails. A uniform magnetic field is directed into the plane of the paper. PQ is given an initial velocity to the right and left to roll. Which statement is **INCORRECT** ?

A. The induced current is in the direction $PQRS$.
B. The magnetic force acting on rod PQ is towards the left.
C. Rod RS starts moving towards the right.
D. Rod PQ moves with a uniform speed.

Q32

32.



The figure shows the location of an isolated charge of size $+Q$. The size (in an arbitrary unit) of the electric field strength is marked at certain points. What is the size (in the same arbitrary unit) of the electric field strength at X and Y ?

	electric field strength at *X*	**electric field strength at *Y***
A.	72	30
B.	72	36
C.	90	30
D.	90	36

Q33

*33. Power is transmitted over long distances at high alternating voltages. Which statements are correct ?

(1) Alternating voltages can be stepped up or down efficiently by transformers.
(2) For a given transmitted power, the current will be reduced if a high voltage is adopted.
(3) The power loss in the transmission cables will be reduced if a high voltage is adopted.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

Q34

34. Which of these is a nuclear fusion reaction ?

A. $^{235}_{92}U + n \rightarrow {}^{144}_{56}Ba + {}^{90}_{36}Kr + 2n$

B. $^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n$

C. $^{14}_{7}N + n \rightarrow {}^{14}_{6}C + {}^{1}_{1}H$

D. $^{238}_{92}U \rightarrow {}^{234}_{90}Th + \alpha$

Q35

*35. On which of the following does the activity of a radioactive source depend ?

(1) the nature of the nuclear radiation emitted by the source
(2) the half-life of the source
(3) the number of active nuclides in the source

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q36

36. Different absorbers are placed in turn between a radioactive source and a Geiger-Muller tube. Three readings are taken for each absorber. The following data are obtained:

Absorber	**Count rate / s^{-1}**		
–	200	205	198
paper	197	202	206
5 mm aluminium	112	108	111
25 mm lead	60	62	58
50 mm lead	34	36	34

What type(s) of radiation does the source emit ?

A. β only
B. γ only
C. β and γ only
D. α , β and γ